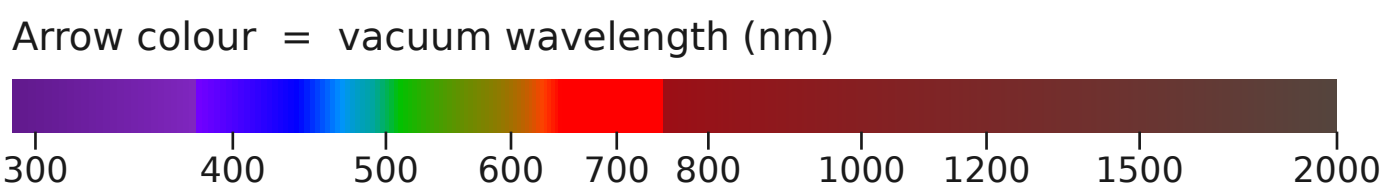


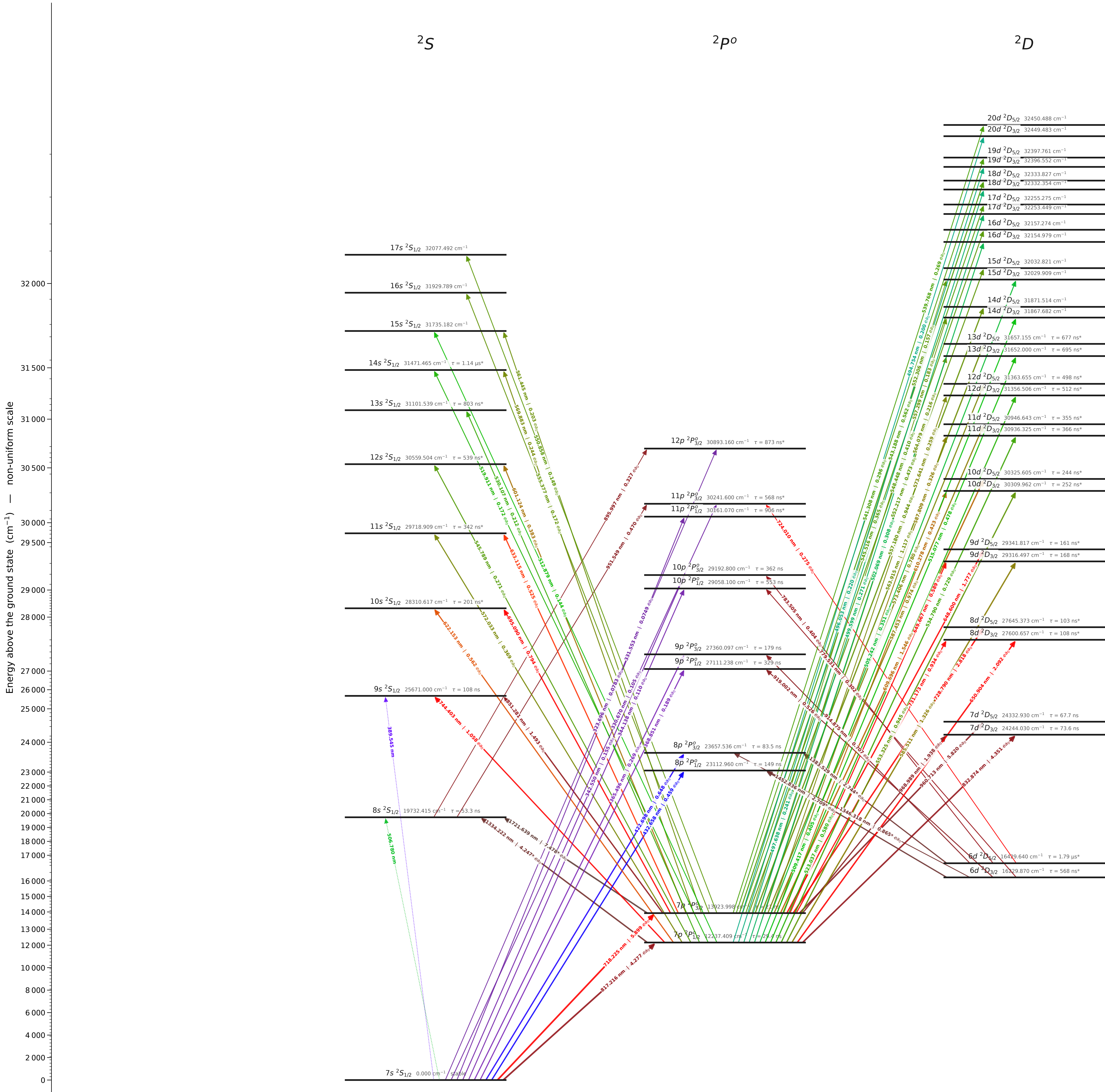
Francium

Arrow label: vacuum wavelength | reduced dipole matrix element $\langle J||e\mathbf{r}||J'\rangle$ in $e a_0$
Matrix elements are measured or NIST values; * = high-accuracy theory, ≈ = model calculation.
Arrow width grows with the Einstein coefficient A. Dotted: forbidden (clock, M1, E2). Dashed: Rydberg excitation, $n = 70$.
Level caption: energy (NIST, or a measurement of the last 20 years where one exists) and measured lifetime (where one exists).



Fr (Fr I). Strongest classified transitions below 2 μm (NIST): 52 levels, 77 lines.

Fr⁺ ionisation limit 32848.87 cm^{-1} = 4.07274 eV (304.424 nm)



Key transitions

Transition	λ vac (nm)	λ air (nm)	ν (THz)	$\langle J e\mathbf{r} J'\rangle$ ($e a_0$)	A (s^{-1})	$\Gamma/2\pi$ up. (MHz)	branch (%)	Use
$7s\ ^2S_{1/2} - 9s\ ^2S_{1/2}$	389.5446	389.4343	769.5972	-	-	1.48	-	two-photon 75-9S spectroscopy of trapped 210Fr
$7s\ ^2S_{1/2} - 8p\ ^2P^o_{3/2}$	422.6983	422.5793	709.23507(3)	0.648	$2.820\text{e}+06$	1.906	45.3	blue line
$7s\ ^2S_{1/2} - 8p\ ^2P^o_{1/2}$	432.6577	432.5361	692.9091	0.459	$2.640\text{e}+06$	1.066	17.4	blue line
$7s\ ^2S_{1/2} - 8s\ ^2S_{1/2}$	506.7803	506.6391	591.5629	-	-	2.986	-	two-photon / parity-violation transition 7S-8S
$7s\ ^2S_{1/2} - 7p\ ^2P^o_{3/2}$	718.2251	718.0272	417.40739(6)	5.899	$4.757\text{e}+07$	7.572	100	D2 line
$7p\ ^2P^o_{1/2} - 9s\ ^2S_{1/2}$	744.4026	744.1976	402.7289	1.058	$2.750\text{e}+06$	1.48	27.2	744.4 nm: decay channel of 9S
$7s\ ^2S_{1/2} - 7p\ ^2P^o_{1/2}$	817.2165	816.9918	366.84584(6)	4.277	$3.396\text{e}+07$	5.404	100	D1 line
$7p\ ^2P^o_{3/2} - 7d\ ^2D_{3/2}$	832.8738	832.6449	359.9494	4.351	$1.660\text{e}+07$	2.162	78.8	832.9 nm: 7P1/2 - 7D3/2
$7p\ ^2P^o_{3/2} - 9s\ ^2S_{1/2}$	851.2810	851.0472	352.1663	1.493	$3.660\text{e}+06$	1.48	34.1	851.3 nm: 7P3/2 - 9S step
$7p\ ^2P^o_{3/2} - 7d\ ^2D_{5/2}$	960.7134	960.4499	312.0519	5.820	$1.290\text{e}+07$	2.351	99.5	960.7 nm: 7P3/2 - 7D5/2
$7p\ ^2P^o_{3/2} - 7d\ ^2D_{3/2}$	968.9892	968.7236	309.3868	1.938	$2.090\text{e}+06$	2.162	19.3	969.0 nm: 7P3/2 - 7D3/2
$7p\ ^2P^o_{3/2} - 8s\ ^2S_{1/2}$	1334.2218	1333.8570	224.6946	4.247*	$7.693\text{e}+06$	2.986	40.9	1.33 μm : second step of the two-photon 7S-8S excitation u...
$7p\ ^2P^o_{3/2} - 8s\ ^2S_{1/2}$	1721.6395	1721.1694	174.1320	7.478*	$1.110\text{e}+07$	2.986	59.1	1.72 μm : decay channel of 8S

λ , ν from NIST level energies unless a measured frequency for this isotope is available. A: measured value or NIST.
 $\Gamma/2\pi$: natural linewidth of the upper level, $1/(2\pi\tau)$.

²¹²Fr hyperfine structure ($I = 5$)

Level	A (MHz)	B (MHz)	Shift of each F level from the centre of gravity (MHz)
$7s\ ^2S_{1/2}$	9064.20(20)	-	F=9/2: -27192.60 F=11/2: +22660.50
$7p\ ^2P^o_{1/2}$	1192.00(20)	-	F=9/2: -3576.00 F=11/2: +2980.00
$7p\ ^2P^o_{3/2}$	97.21(10)	-26.00(20)	F=7/2: -886.16 F=9/2: -431.81 F=11/2: +107.61 F=13/2: +722.58
$8p\ ^2P^o_{3/2}$	373.00(10)	-	F=9/2: -1119.00 F=11/2: +932.50
$8p\ ^2P^o_{1/2}$	32.80(10)	-7.7(9)	F=7/2: -298.54 F=9/2: -145.93 F=11/2: +35.88 F=13/2: +244.07
$8d\ ^2D_{3/2}$	13.0(6)	-	F=7/2: -117.00 F=9/2: -58.50 F=11/2: +13.00 F=13/2: +97.50
$8d\ ^2D_{5/2}$	-7.2(6)	-	F=5/2: +108.00 F=7/2: +82.80 F=9/2: +50.40 F=11/2: +10.80 F=13/2: -36.00 F=15/2: -90.00
$10s\ ^2S_{1/2}$	401(5)	-	F=9/2: -1203.00 F=11/2: +1002.50
$9d\ ^2D_{3/2}$	7.1(7)	-	F=7/2: -63.90 F=9/2: -31.95 F=11/2: +7.10 F=13/2: +53.25
$9d\ ^2D_{5/2}$	-3.6(4)	-	F=7/2: +54.00 F=9/2: +41.40 F=11/2: +25.20 F=13/2: +5.40 F=15/2: -18.00 F=17/2: -45.00
$11s\ ^2S_{1/2}$	225(3)	-	F=9/2: -675.00 F=11/2: +562.50

Measured A, B constants; sources in the data file.

Isotope shifts

Transition	λ vac (nm)	Isotopes	Shift (MHz)
$7s\ ^2S_{1/2} - 8p\ ^2P^o_{3/2}$	422.698	213-212	-1615.0(3)
$7s\ ^2S_{1/2} - 8p\ ^2P^o_{3/2}$	422.698	220-212	-20564.0(7)
$7s\ ^2S_{1/2} - 8p\ ^2P^o_{3/2}$	422.698	221-212	-23298.0(8)
$7s\ ^2S_{1/2} - 8s\ ^2S_{1/2}$	506.780	213-208	-5123(6)
$7s\ ^2S_{1/2} - 8s\ ^2S_{1/2}$	506.780	213-209	-3678(6)
$7s\ ^2S_{1/2} - 8s\ ^2S_{1/2}$	506.780	213-210	-3274(6)
$7s\ ^2S_{1/2} - 8s\ ^2S_{1/2}$	506.780	213-211	-1958(6)
$7s\ ^2S_{1/2} - 7p\ ^2P^o_{3/2}$	718.225	207-212	5239(4)
$7s\ ^2S_{1/2} - 7p\ ^2P^o_{3/2}$	718.225	208-212	5003(3)
$7s\ ^2S_{1/2} - 7p\ ^2P^o_{3/2}$	718.225	209-212	3133.0(20)
$7s\ ^2S_{1/2} - 7p\ ^2P^o_{3/2}$	718.225	210-212	2603.0(10)
$7s\ ^2S_{1/2} - 7p\ ^2P^o_{3/2}$	718.225	211-212	901(3)
$7s\ ^2S_{1/2} - 7p\ ^2P^o_{3/2}$	718.225	222-212	-26262(3)
$7s\ ^2S_{1/2} - 7p\ ^2P^o_{3/2}$	718.225	223-212	-27922.0(20)
$7s\ ^2S_{1/2} - 7p\ ^2P^o_{3/2}$	718.225	224-212	-30891.0(10)
$7s\ ^2S_{1/2} - 7p\ ^2P^o_{3/2}$	718.225	225-212	-32297(6)
$7s\ ^2S_{1/2} - 7p\ ^2P^o_{3/2}$	718.225	226-212	-34401.0(10)
$7s\ ^2S_{1/2} - 7p\ ^2P^o_{3/2}$	718.225	227-212	-38352.0(20)
$7s\ ^2S_{1/2} - 7p\ ^2P^o_{3/2}$	718.225	228-212	-40077(5)
$7s\ ^2S_{1/2} - 7p\ ^2P^o_{3/2}$	718.225	221-212	-23570.0(20)
$7s\ ^2S_{1/2} - 7p\ ^2P^o_{3/2}$	718.225	213-212	-1641(3)
$7s\ ^2S_{1/2} - 7p\ ^2P^o_{3/2}$	718.225	220-212	-20806.8(5)

Shift = ν (first isotope) - ν (second isotope).